



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

19-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	D	D	D	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C	A,B	B,D	A,C	A,C	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	4.00	2.00	161.20 or 161.40	4.00	1.0

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	D	B	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,C,D	A,B,D	B,C,D	C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	15.00	3.00	3.00	2.00	6.00	7.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	B	B	C	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,D	B,D	B,C,D	A,B,C	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	25.00	5.20	2.00	8.00	6.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

5. Ans (B)

$$B \propto \frac{1}{A}$$

$$\frac{BNA}{CR} = \frac{Q}{LR}$$

Voltage sensitivity

6. Ans (D)

$$x = -a \sin(2\omega t) = -2a \sin \omega t \cos \omega t$$

$$x = -2a \sin \omega t \sqrt{1 - \sin^2(\omega t)}$$

$$x^2 = 4a^2 \left(\frac{y}{2a} \right)^2 \left[1 - \frac{y^2}{4a^2} \right]$$

$$x^2 = y^2 \left[1 - \frac{y^2}{4a^2} \right]$$

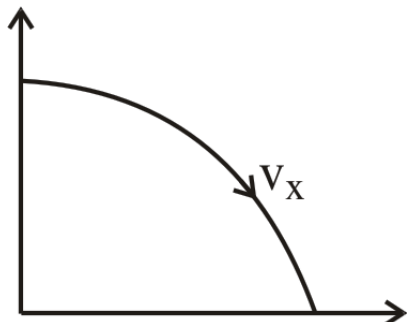
PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (A,B)

$$\frac{1}{2}m4v_0^2 - \frac{1}{2}mv_0^2 = q \in 2a$$

$$\frac{3mv_0^2}{4a} = \in$$



$$q \left[(v_x \hat{i} - v_y \hat{j}) \times B_0 (-\hat{k}) \right] + q \in_0 \hat{i}$$

$$q[v_x B] \hat{j} + qv_y B_0 \hat{i} + q \in_0 \hat{i}$$

By impulse momentum theorem

$$\int qv_y B_0 dt + q \in_0 t = -mv_0$$

$$-qB_0 a + q \in_0 t = -mv_0$$

$$t = \frac{qB_0 a - mv_0}{q \in_0}$$

9. Ans (B,D)

$$L_1 + L_2 + 2m = 120 \text{ mH}$$

$$L_1 + L_2 - 2m = 60 \text{ mH}$$

$$3L + 2m = 80 \quad \dots(i)$$

$$3K - 2m = 40 \quad \dots(ii)$$

$$(i) - (ii)$$

$$6L = 20$$

$$M = 1 < \sqrt{L_1 L_2}$$

10. Ans (A,C)

$$\text{Intensity} = \frac{20}{4\pi} \text{ watt}$$

$$I = \frac{1}{2} \in_0 E^2 c$$

$$\frac{20}{4\pi} = \frac{1}{2} \in_0 E^2 c$$

$$40 \times 9 \times 10^9 = E^2$$

$$\frac{36 \times 10^{10}}{3 \times 10^8} = E^2 \Rightarrow 34.7 \text{ V/m}$$

$$B = \frac{6 \times 10^5}{3 \times 10^8} = 2 \times 10^{-3}$$

12. Ans (A,B)

$$W = \sqrt{\frac{K}{m}} = \sqrt{\frac{6 \text{ N/m}}{3}} \times 4 = 2$$

We can find max extension by applying work

energy theorem

$$0 - \frac{1}{x} \times \frac{2}{3} \times 4 = -\frac{1}{2} K x^2$$

$$x^2 = \frac{4}{9} \Rightarrow \frac{2}{3} \text{ m}$$

maximum force on block $Kx = 4\text{N}$

Therefore ACC can be 1 to 4 m/s².

PART-1 : PHYSICS

SECTION-II

4. Ans (161.20 or 161.40)

Power loss across resistor

$$P = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{\left(R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right)} \times R$$

$$\omega L = 2 \times \frac{22}{7} \times 50 \times \frac{35}{44} = 250 \Omega$$

$$\frac{1}{\omega L} = \frac{10^6}{2 \times \frac{22}{7} \times 50 \times \frac{700}{22}} = 100 \Omega$$

$$P = \frac{(220)^2}{(150)^2 \times 2} \times 150 = \frac{(220)^2}{150 \times 2} = 161.33 \text{ W}$$

5. Ans (4.00)

$$|\vec{E}| \times 2\pi r = - \frac{\partial B}{\partial t} \times \pi r^2$$

$$\Rightarrow |\vec{E}| = \frac{r}{2} \frac{\partial B}{\partial t} = \frac{2 \times 10^{-2}}{2} \times 0.4 = 4 \text{ mV/m}$$

6. Ans (1.0)

Induced emf $e = B\ell v$

$$= (0.2 \times 10^{-4}) \times 1 \text{ m} \times 180 \times \frac{1000}{60 \times 60} \text{ V} = 1 \text{ mV}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (B)

$$\text{Osmotic pressure} = \frac{w}{M \times V} \times R \times T \quad \text{or}$$

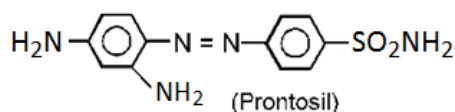
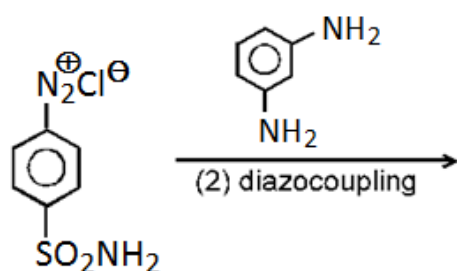
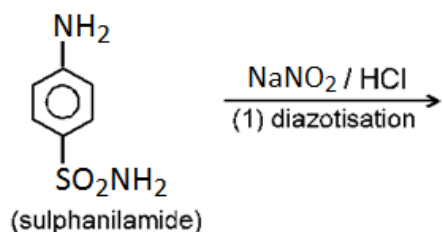
$$M = \frac{w}{P \times V} \times R \times T$$

$$M = \frac{10}{5 \times 10^{-3} \times 1} \times 0.0821 \times 298 = 48931.6 \text{ g/mol}$$

Each glucose unit in starch has molecular mass
= 180 - 18 = 162 g/mol

$$\text{The number of glucose units in the sample of starch} = \frac{48931.6}{162} = 302$$

5. Ans (D)



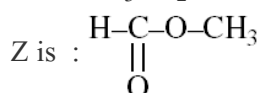
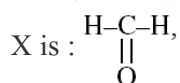
6. Ans (C)

(iii), (iv), (v), (vii)

PART-2 : CHEMISTRY

SECTION-I (ii)

10. Ans (B,C,D)



PART-2 : CHEMISTRY

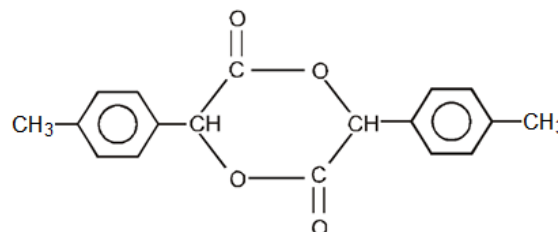
SECTION-II

1. Ans (15.00)

(x = 6, y = 1, z = 8)

2. Ans (3.00)

Product R is



4. Ans (2.00)

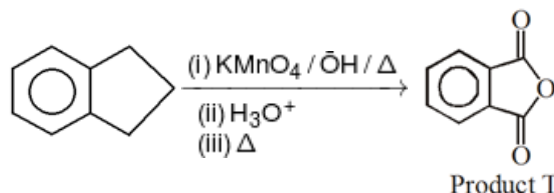
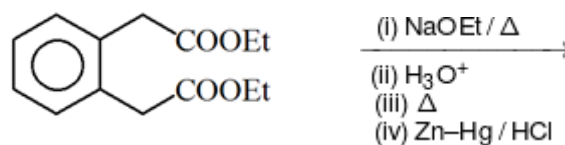
y = 2

Serotonin, Histamine

5. Ans (6.00)

except carboxylic acid all compound give positive NaHCO_3 test.

6. Ans (7.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

A : Event that article came from lot A

B : Article came from B

D : Article from mixed lot is defective

$$P(A) = \frac{3}{7}, P(B) = \frac{4}{7}$$

$$P(D) = P(D \cap A) + P(D \cap B)$$

$$= P(A) \cdot P\left(\frac{D}{A}\right) + P(B) \cdot P\left(\frac{D}{B}\right) =$$

$$\frac{3}{7} \times \frac{3}{8} + \frac{4}{7} \times \frac{5}{8} = \frac{29}{56}$$

2. **Ans (A)**

$$a = 2b = 2c$$

∴ no. of ordered triples (a, b, c) satisfying are 3
((2, 1, 1), (4, 2, 2), (6, 3, 3))

2 out of these 3 ((2, 1, 1) and (4, 2, 2)) lie inside the tetrahedron

$$\therefore \text{Required probability} = \frac{2}{3} = \frac{6}{\lambda} \Rightarrow \lambda = 9$$

3. **Ans (B)**

P(2 aces are drawn in exactly n drawn) =

P(exactly 1 ace in n - 1 draws) × P(second ace in nth draw)

$$= \frac{{}^{48}C_{n-2} \cdot {}^4C_1}{{}^{52}C_{n-1}} \times \frac{{}^3C_1}{53-n}$$

$$= \frac{(n-52)(n-51)(n-1)}{13 \cdot 17 \cdot 50 \cdot 49} = \frac{1}{k}(n-a)(n-b)(n-c)$$

$$\Rightarrow a = 52, b = 51, c = 1$$

$$\therefore a - b - c = 0$$

4. **Ans (B)**

AAAAA, BBBB, KKKKK, OOOO

Each get 4 fruits

P Q R S T are five person

$${}^5C_2 [{}^4C_2 \times 2! ({}^3C_2 \times 2! \times 2!)] = 1440$$

5. **Ans (C)**

$$\begin{aligned} & \sqrt{\sin 4x + 4\cos^2 x} + \sqrt{\cos^4 x + 4\sin^2 x} \\ &= \sqrt{\sin^4 x - 4\sin^2 x + 4} + \sqrt{\cos^4 x + 4 - 9\cos^2 x} \\ &= 2 - \sin^2 x + 2 - \cos^2 x = 3 \end{aligned}$$

$$\Rightarrow \sqrt{1 + \frac{3 \cos 2x}{3}} = \sqrt{1 + \cos 2x}$$

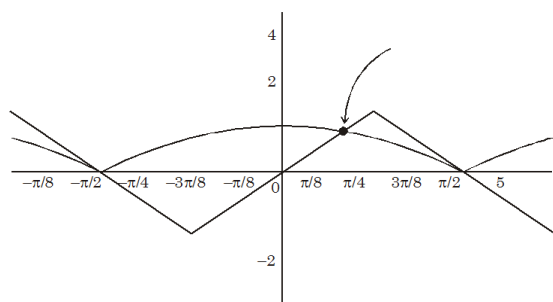
$$\Rightarrow |\cos x| = \sin^{-1}(2x)$$

$$f(x) = |\cos x| - \sin^{-1}(2x)$$

$$f\left(\frac{\pi}{8}\right) = \left|\cos \frac{\pi}{8}\right| - \sin^{-1}\left(\sin \frac{\pi}{4}\right) = \cos \frac{\pi}{8} - \frac{\pi}{4} > 0$$

$$f\left(\frac{\pi}{6}\right) = \left|\cos \frac{\pi}{6}\right| - \sin^{-1}\left(\sin \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{3} < 0$$

⇒ one root exists between $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$



6. **Ans (D)**

$$\begin{aligned} & \log_3(\sqrt{x} + |\sqrt{x} - 1|) = \log_9(4\sqrt{x} - 3 + 4|\sqrt{x} - 1|) \\ & \Rightarrow \log_3(\sqrt{x} + |x - 1|)^2 = \log_3(4\sqrt{x} - 3 + 4|\sqrt{x} - 1|) \\ & \Rightarrow (\sqrt{x} + |\sqrt{x} - 1|)^2 = 4\sqrt{x} - 3 + 4|\sqrt{x} - 1| \\ & x + (\sqrt{x} - 1)^2 + 2\sqrt{x}|\sqrt{x} - 1| = 4\sqrt{x} + 4|\sqrt{x} - 1| - 3 \\ & \Rightarrow x - 3\sqrt{x} + 2 = |\sqrt{x} - 1|(2 - \sqrt{x}) \\ & (\sqrt{x} - 1)(\sqrt{x} - 2) + |\sqrt{x} - 1|(\sqrt{x} - 2) = 0 \end{aligned}$$

If :

$$\sqrt{x} - 1 \geq 0 \Rightarrow x \geq 1$$

$$2(\sqrt{x} - 1)(\sqrt{x} - 2) = 0$$

$$\Rightarrow \sqrt{x} = 1 \text{ or } \sqrt{x} = 2$$

$$\Rightarrow x = 1 \text{ or } x = 4$$

If :

$$\sqrt{x} - 1 < 0 \Rightarrow x < 1$$

$$\Rightarrow (\sqrt{x} - 1)(\sqrt{x} - 2) - (\sqrt{x} - 1)(\sqrt{x} - 2) = 0$$

$$\Rightarrow 0 = 0$$

$$\Rightarrow \text{true for all } x < 1$$

$$\Rightarrow 0 \leq x < 1$$

$$\Rightarrow \text{common solution } [0, 1] \cup \{4\}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A,B,C)**

Let a, b be the no. of shots A, B respectively take until they hit the target.

Option (A) :

$$S = \sum_{n=1}^{\infty} (P(a = n)) \times (P(b > n))$$

for $P(a = n)$, A must miss their first $n - 1$ shots and hit on the nth shot.

$$\therefore P(a = n) = \left(1 - \frac{3}{5}\right)^{n-1} \times \frac{3}{5} = \frac{3}{5} \times \left(\frac{2}{5}\right)^{n-1}$$

&

$$P(b > n) = \sum_{k=n+1}^{\infty} P(b = k) = \sum_{k=n+1}^{\infty} \left(\frac{2}{7}\right)^{k-1} \times \frac{5}{7} = \left(\frac{2}{7}\right)^n$$

$$\therefore S = \sum_{n=1}^{\infty} \left(\frac{3}{5} \times \frac{2}{7}\right)^{n-1} \times \frac{3}{5} \times \frac{5}{7} = \frac{3}{2} \times \frac{\frac{4}{35}}{1 - \frac{4}{35}} = \frac{6}{31}$$

Option C :

Probability that A, B require same number of shots will be

$$\sum_{n=1}^{\infty} (P(A=n)) \times (P(B=n)) = \sum_{n=1}^{\infty} \left(\frac{4}{35}\right)^{n-1} \times \frac{3}{5} \times \frac{5}{7}$$

$$= \frac{3}{7} \times \frac{1}{1 - \frac{4}{35}} = \frac{15}{31}$$

Option B :

Probability that B require less shots than A will

$$be 1 - \frac{6}{31} - \frac{15}{31} = \frac{10}{31}$$

8. **Ans (A,D)**

$$B_1 \rightarrow 2\ell - 1, B_2 \rightarrow 2m + 2, B_3 \rightarrow 2h + 3, B_4 \rightarrow$$

$2p$ (general term) even

$$\therefore 2\ell - 1 + 2m + 2 + 2h + 3 + 2p = 50$$

$$\Rightarrow 2\ell + 2m + 2h + 2p = 46$$

$$\Rightarrow \ell + m + n + p = 23 ; \ell, m, n, p \geq 1$$

$$\Rightarrow \ell' + m' + n' + p' = 19 ; \ell', m', n', p' \geq 0$$

$$\therefore \text{total number of ways of distribution} = {}^{22}C_3$$

for option C or D.

no. of ways of B_4 receives not more than 14 marbles

= total no. of ways of - no. of ways of distribution when B_4 receives 16 or more marbles

$$\ell + m + n + p = 23$$

$$1 \quad 1 \quad 1 \quad 8$$

$$\ell' + m' + n' + p' = 12,$$

$$\ell', m', n', p' \geq 0$$

$$\text{no. of ways} = {}^{15}C_3$$

$$\text{Ans} = {}^{22}C_3 - {}^{15}C_3 = 1085$$

9. **Ans (B,D)**

Let $p = \cos x \sin y \cos z$. Because $\frac{\pi}{2} \geq y \geq z$,

$\sin(y - z) \geq 0$. By the product-to-sum formulas, we have

$$p = \frac{1}{2} \cos x [\sin(y + z) + \sin(y - z)] \geq \frac{1}{2} \cos x \sin(y + z) = \frac{1}{2} \cos^2 x$$

$$\text{Note that } x = \frac{\pi}{2} - (y + z) \leq \frac{\pi}{2} - 2 \cdot \frac{\pi}{12} = \frac{\pi}{3}.$$

Hence the minimum value of p is $\frac{1}{2} \cos^2 \frac{\pi}{3} = \frac{1}{8}$,

obtained when $x = \frac{\pi}{3}$ and $y = z = \frac{\pi}{12}$.

$$p = \frac{1}{2} \cos z [\sin(x + y) - \sin(x - y)] \leq \frac{1}{2} \cos^2 z$$

by noting that $\sin(x - y) \geq 0$ and $\sin(x + y) = \cos z$.

By the double-angle formulas, we deduce that

$$p \leq \frac{1}{4} (1 + \cos 2z) \leq \frac{1}{4} \left(1 + \cos \frac{\pi}{6}\right) = \frac{2 + \sqrt{3}}{8}$$

This maximum value is obtained if and only if

$$x = y = \frac{5\pi}{24} \text{ and } z = \frac{\pi}{12}$$

10. **Ans (B,C,D)**

$$S_n = 6 + 17 + 34 + 57 + \dots + T_n$$

$$S_n = 6 + 17 + 34 + 57 + \dots + T_{n-1} + T_n$$

$$0 = 6 + 11 + 17 + 23 + \dots + (T_n - T_{n-1}) - T_n$$

$$T_n = 11 + 17 + 23 + \dots + (T_n - T_{n-1})$$

$$= 6 + \frac{n-1}{2} (2 \times 11 + (n-1-1)6)$$

$$T_n = 6 + (n-1)(3n+5)$$

$$T_n = 3x^2 + 2x + 1 \equiv (\log_2 a)n^2 + (\log_3(b-a))n + \log_4 c$$

$$\therefore \log_2 a = 3 \Rightarrow a = 8; \log_3(b-a) = 2$$

$$\Rightarrow b - a = 9 \Rightarrow b = 17$$

$$\log_4 c = 1 \Rightarrow c = 4$$

Now, verify the options

11. Ans (A,B,C)

$$(A) \sum_{r=0}^n {}^nC_r + 2 \sum_{r=0}^n \frac{{}^nC_r}{(r+1)}$$

$$2^n + 2 \left(\frac{2^{(n+1)} - 1}{(n+1)} \right)$$

$$(B) \lambda = 950 = (10-1)^{50}$$

$$= {}^{50}C_0 10^5 - {}^{50}C_1 10^4 + \dots + 1$$

$$= \text{last digit is } 1$$

$$(C) (2-1) + (2^2-1) + (2^3-1) + \dots + (2^{20}-1)$$

$$2(2^{20}-1) - 20 = 2^{21} - 22$$

$$(D) \sum_{r=1}^{10} (r-1)! - r! = (11)! - 11$$

$$= \text{even} - 1 = \text{odd}$$

12. Ans (A,D)

$$\text{Avg} = \frac{2 \sin 2^\circ + 4 \sin 4^\circ + 6 \sin 6^\circ + \dots + 180 \sin 180^\circ}{90}$$

$$A = \frac{1}{90} \{2 \sin 2^\circ + 4 \sin 4^\circ + 6 \sin 6^\circ + \dots + 178 \sin 178^\circ\}$$

$$A(\sin 1^\circ) = \frac{1}{90} \{2 \sin 2^\circ \sin 1^\circ + 4 \sin 4^\circ \sin 1^\circ + 6 \sin 6^\circ \sin 1^\circ + \dots\}$$

$$[2 \sin 2k^\circ \sin 1^\circ = \cos(2k-1) - \cos(2k+1)]$$

$\Rightarrow$

$$A(\sin 1^\circ) = \frac{1}{90} \{(\cos 1^\circ - \cos 3^\circ) + 2(\cos 3^\circ - \cos 5^\circ) + \dots + 89(\cos 177^\circ - \cos 179^\circ)\}$$

$$A(\sin 1^\circ) = \frac{1}{90} \{\cos 1^\circ + \cos 3^\circ + \cos 5^\circ + \dots + \cos 177^\circ - 89 \cos 179^\circ\}$$

$$A(\sin 1^\circ) = (\cos 1^\circ + \cos 3^\circ + \cos 177^\circ) + \dots + (\cos 89^\circ + \cos 91^\circ) + 89 \cos 1^\circ$$

$$\therefore A(\sin 1^\circ) = \frac{1}{90} (90 \cos 1^\circ)$$

$$\Rightarrow \text{Avg} = \cot 1^\circ$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4.00)

$${}^9C_3 \times {}^3C_2 \times \frac{5!}{2!2!1!}$$

2. Ans (25.00)

$$(\sin x + \cos x)(\tan^2 x - (1 + \sqrt{3}) \tan x + 3 + \sqrt{3}) = 3 |\sin x + \cos x|$$

Case I :

$$\sin x + \cos x \geq 0$$

$$(s+c)(t^2 - (1 + \sqrt{3})t + \sqrt{3}) = 0$$

$$\Rightarrow t = -1, 1, \sqrt{3}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{7\pi}{4}$$

Case II :

$$s+c < 0$$

$$(s+c)(t^2 - (1 + \sqrt{3})t + 6 + \sqrt{3}) = 0$$

$$\Rightarrow t = -1 \quad \text{No solution}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\Rightarrow \text{sum of solutions is } \frac{37\pi}{12}$$

$$|p-q| = 25$$

3. Ans (5.20)

$$t_1 = \frac{\sin x}{\cos 3x}, t_2 = \frac{\sin 3x}{\cos 9x}, t_3 = \frac{\sin 9x}{\cos 27x}$$

$$t_1 = \frac{\sin x \cdot 2 \cos x}{\cos 3x \cdot 2 \cos x} = \frac{\sin 2x}{2 \cos x \cos 3x} = \frac{\sin(3x-x)}{2 \cos x \cos 3x}$$

$$= \frac{1}{2} (\tan 3x - \tan x)$$

$$\text{Similarly, } t_2 = \frac{1}{2} (\tan 9x - \tan 3x)$$

$$t_3 = \frac{1}{2} (\tan 27x - \tan 9x)$$

$$t_1 + t_2 + t_3 = 0 \Rightarrow \frac{1}{2} (\tan 27x - \tan x) = 0$$

$$\Rightarrow \tan 27x = \tan x$$

$$\Rightarrow 27x = n\pi + x$$

$$\Rightarrow 26x = n\pi \Rightarrow x = \frac{n\pi}{26}$$

$$n = 1 \Rightarrow x = \frac{\pi}{26} \quad \therefore k = 26$$

4. Ans (2.00)

$$x+y+z=\theta \Rightarrow \cos x + \cos y + \cos z = 2 \cos \theta$$

$$\& \sin x + \sin y + \sin z = 2 \sin \theta$$

$$\& \cos(x+y) + \cos(y+z) + \cos(z+x) = ?$$

$$= \cos(\theta-2) + \cos(\theta-x) + \cos(\theta-y)$$

$$= \cos \theta \cos z + \sin \theta \sin z + \cos \theta \cos x + \sin \theta \sin x + \cos \theta \cos y + \sin \theta \sin y$$

$$= \cos \theta (\cos x + \cos y + \cos z) + \sin \theta (\sin x + \sin y + \sin z) = 2 \cos 2\theta + 2 \sin^2 \theta = 2$$

5. Ans (8.00)

Replace $x \rightarrow \frac{2}{x}$

$$\left(\frac{8}{x^2} + \frac{6}{x} + 4 \right)^{10} = \sum_{r=0}^{20} a_r \left(\frac{2}{x} \right)^r$$

$$\Rightarrow \frac{2^{10}(2x^2 + 3x + 4)^{10}}{x^{20}} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{-r}$$

$$\Rightarrow 2^{10}(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$$\Rightarrow 2^{10} \sum_{r=0}^{20} a_r \cdot x^r = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$\therefore$ coefficient of x^7

$$2^{10} a_7 = a_{13} 2^{13}$$

$$\text{p } \frac{a_7}{a_{13}} = 2^3 = 8$$

6. Ans (6.00)

$$T_{r+1} = {}^nC_r 2^r x^r \Rightarrow a_k = {}^nC_k 2^k$$

$$\sum_{k=0}^n (3k+1) {}^nC_k \cdot a_k = 3 \sum_{k=0}^n k {}^nC_k \cdot 2^k + \sum_{k=0}^n {}^nC_k \cdot 2^k$$

$$= 3 \sum_{k=0}^n n \cdot n^{-1} {}^nC_{k-1} \cdot 2^k + (1+2)^n$$

$$= 3n \sum_{k=0}^n n^{-1} {}^nC_{k-1} 2^{k+1} + 3^n$$

$$= 2.3n(1+2)^{n-1} + 3^n$$

$$= 2n \cdot 3^n + 3^n = (2n+1)3^n$$

$$p = 2, q = 1, r = 3$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

19-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,D	B,C	A,C,D	A,B,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.08	1.88	2.09 to 2.10	2280.00	1.50 to 1.56	5
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	1	5	6	2	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,C,D	A	A,C,D	A,B	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	116.64	25.33	3.00	10.00	1.50	495.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	8	2	6	4	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	C	C	A,C	B,D	B,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.05	2.00	0.00	6.00	19.00	0.36
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	5	3	2	6	4

HINT – SHEET

PART-1 : PHYSICS

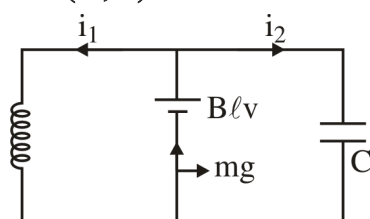
SECTION-I

1. Ans (B,C)

$$A = \frac{5}{100}(\text{m}) = 5 \text{ cm}$$

Position of ring at a phase angle of 37° and position of wall at a phase of 53° therefore time period $\pi\sqrt{\frac{M}{K}}$.

3. Ans (B,C)



By applying K.V.L. in the loop

$$B\ell v = L \frac{di_1}{dt} \quad \dots(1)$$

$$i_1 = \frac{B\ell}{L} \int v dt \Rightarrow LI_1 = B\ell x \Rightarrow I_1 = \frac{B\ell x}{L}$$

$$C(B\ell v) = Q \text{ (in capacitor)}$$

$$i_2 = B\ell Ca$$

Force equation on the rod

$$mg - B(\ell_1 + \ell_2)\ell = ma$$

$$mg - Bi_1\ell = ma + Bi_2\ell$$

$$mg - B\ell \left(\frac{B\ell x}{L} \right) = ma + B^2\ell^2 Ca$$

$$\frac{1}{m + B^2\ell^2 C} \left[mg - \frac{B^2\ell^2 x}{L} \right] = a$$

(represent eq. of SHM)

$$\omega^2 = \frac{B^2\ell^2}{2ML} \Rightarrow \omega = \frac{B\ell}{\sqrt{2m\ell}}$$

5. Ans (A,B,D)

By K.V.L.

$$\frac{L di}{dt} - iR + \frac{1}{2} B \omega a = 0$$

$$\frac{L di}{dt} + iR = \frac{1}{2} \left(\frac{r_0 N i}{\ell} \right) \omega a^2$$

$$\frac{di}{dt} = \frac{1}{L} \left[\frac{1}{2} \left(\frac{r_0 N i}{\ell} \right) \omega a^2 - iR \right]$$

$$\frac{di}{dt} > 0$$

$$\frac{1}{2} \frac{r_0 N i \omega a^2}{\ell} > iR$$

$$\omega > \frac{2R\ell}{r_0 N a^2}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (4.08)

$$I = \frac{25K\omega}{\pi R^2} = \frac{1}{2} \epsilon_0 E^2 C$$

$$E^2 = \frac{50 \times 10^3}{\pi \epsilon_0 \times 36 \times 10^{-4} \times 3 \times 10^8}$$

$$\epsilon^2 = \frac{50 \times 10^3 \times 36 \times 10^9}{36 \times 10^{-4} \times 3 \times 10^8} \Rightarrow \frac{50}{3} \times 10^8$$

$$E^2 = 16 \times 10^8$$

2. Ans (1.88)

$$Kx_S + Kx_B = m\omega^2 x_B$$

$$\frac{Kx}{4} + \frac{K21x}{4} = \frac{m\omega^2 21x}{4}$$

$$\frac{22Kx}{4} = \frac{m\omega^2 21x}{4}$$

$$\omega = \sqrt{\frac{22K}{21m}}$$

3. Ans (2.09 to 2.10)

$$\int_A^B B \cdot d\ell = \frac{\mu_0 i}{2\pi R} R \frac{\pi}{6}$$

$$\frac{\mu_0 i}{12} \Rightarrow \frac{\mu_0}{6} \Rightarrow \frac{4\pi}{6} \times 10^{-8} \Rightarrow \frac{6.28}{3} = 2.09$$

4. Ans (2280.00)

$$\frac{1200 \times 10^3}{4000} = 300 \text{ A}$$

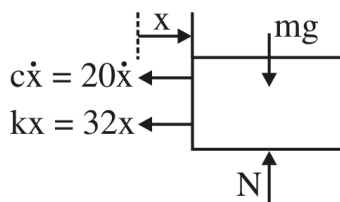
$$\text{Line loss of power} = (300)^2 \Omega$$

$$= (300)^2 \times 15 = 9 \times 10^4 \times 12 = 1080 \text{ KW}$$

$$\text{Power plant power} = 1200 \text{ KW} + 1080 \text{ KW}$$

$$= 2280 \text{ KW}$$

5. Ans (1.50 to 1.56)



We must first determine whether the system is underdamped, critically damped, or overdamped.

For that purpose, we compute the damping ratio ξ .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{32}{8}} = 2 \text{ rad/s}$$

$$\xi = \frac{c}{2m\omega_n} = \frac{20}{2(8)(2)} = 0.625$$

Since $\xi < 1$, the system is underdamped. The

damped natural frequency is

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 2 \sqrt{1 - (0.625)^2} = 1.561 \text{ rad/s}$$

6. Ans (5)

Applying conservation of flux,

$$L_i I_i = L_f I_f$$

$$\Rightarrow 5 \times 1 = 1 \times I_f \text{ So } I_f = 5 \text{ A}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (1)**

$$X_L = \omega L = \frac{5}{\sqrt{3}}(3.6) = 6\sqrt{3}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{\frac{5}{\sqrt{3}}(0.1)} = 2\sqrt{3}$$

$$R = 4a$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = 8 \text{ ohm}$$

e.m.f. leads current by 60°

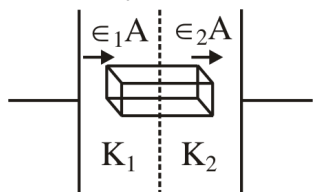
$$I = 2 \sin\left(\frac{5}{\sqrt{3}}t - 60^\circ\right)$$

$$V_C = 4\sqrt{3} \sin\left(\frac{5}{3}t - 60^\circ - 90^\circ\right)$$

$$C = \frac{\epsilon_0 A}{d} \Rightarrow 0.1 = \frac{\epsilon_0 \times 10^{-4}}{d} \Rightarrow d = \epsilon_0 10^{-3}$$

Electric field in capacitor

$$\frac{V_C}{d} = \frac{4\sqrt{3}}{\epsilon_0 \times 10^{-3}} \sin$$



by applying Gauss Law

$$\left(\frac{K_2 - K_1}{K_1 K_2}\right) (\epsilon_2 - \epsilon_1) A = \frac{Q_{in}}{\epsilon_0}$$

$$\frac{\sigma}{\epsilon_0} = \left[\frac{2}{4}\right] = \frac{4\sqrt{3}}{\epsilon_0 \times 10^{-3}} \sin\left(\frac{5}{\sqrt{3}} \times \frac{7\pi}{10\sqrt{3}} = 150^\circ\right)$$

2. **Ans (1)**

$$\frac{M}{L} = \frac{Q}{2M} \Rightarrow \vec{M} \text{ (magnitude of cone)} = \left(\frac{\pi R^2}{2} i\right)$$

$$\text{Torque of magnetic force } MB \sin \theta \Rightarrow \frac{\pi R^2 i B}{2}$$

$$\tau_B > \tau_{mg} \Rightarrow \frac{\pi R^2 i B}{2} = mgR$$

$$\Rightarrow i = 2mgBR$$

3. **Ans (5)**

$$X_L = \omega L \Rightarrow 100 \times 80 \times 10^{-3} = 82$$

$$X_C = \frac{1}{\omega C} \Rightarrow \frac{1}{100 \times 2.5 \times 10^{-3}} \Rightarrow 4\Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{8i} + \frac{1}{4i} \Rightarrow i \left[\frac{1}{8}\right] \Rightarrow R_{eq} = -8i$$

$$Z_{eq} = 6i - 8i \Rightarrow \sqrt{36 + 64} \Rightarrow 10$$

4. **Ans (6)**

$$\frac{1}{2} \times 8 \times x = \frac{\phi_2 - \phi_1}{2}$$

Area of $i-t$ graph is charge $3C$.

$$Q = \frac{\phi_2 - \phi_1}{R} \Rightarrow 6 \text{ weber}$$

5. **Ans (2)**

When velocity is maximum then net force acting on the conductor is zero.

6. **Ans (2)**

Initially when the magnetic field is building up, impulses of the force of interaction between the currents induced in the jumpers due to the time varying magnetic field and the magnetic field provide each of the jumpers an inward momentum. Thereafter, each jumper will eventually stop by losing its momentum due to the impulse of force of interaction between the static magnetic field and the current caused by motional emf in the jumper.

$$\frac{Blx_0}{\Delta t} = I \times 2R \Rightarrow I = \frac{Blx_0}{2R\Delta t}$$

$$\text{Change in momentum } P = IIB\Delta t = B^2 l^2 x_0 / 2R$$

Now at an instant let velocity of conductor is v , so emf is induced in the conductor.

$$I_{in} = \frac{2Blv}{2R}, F = IIB = \frac{B^2 l^2 v}{R}$$

$$\int F dt = \frac{B^2 l^2}{R} \int v dt = P$$

$$\Rightarrow \int v dt = x_0 / 2$$

PART-2 : CHEMISTRY

SECTION-I

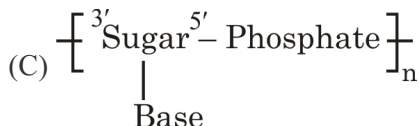
1. **Ans (A,B,C,D)**

NCERT text book

2. **Ans (A,C,D)**

(A) Paired base sequence from 5' to 3' will be TGACACTG.

(B) Bases are derived from pyrimidine.

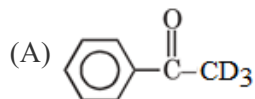
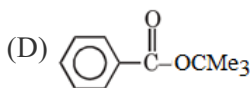
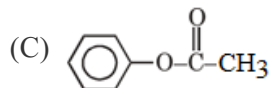
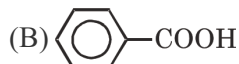
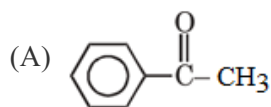


(D) A:::T, G:::C

3. **Ans (A)**

NCERT Theory

4. **Ans (A,C,D)**



(B) Dimerization phenomenon, $i < 1$.

(C) Perkin reaction
(148 amu)

(D) One is BACl_2 and another is BAI_3

5. **Ans (A,B)**

(A) Reduction of amides

(B) Carbylamine reaction reduction of isocyanides

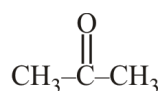
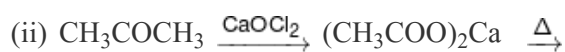
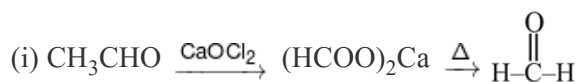
(C) Imine formation takes place

(D) Nucleophilic addition not possible

6. **Ans (A,D)**

(A) Hydration depends on extent of partial positive charge & steric hinderance.

(B) Haloform reaction



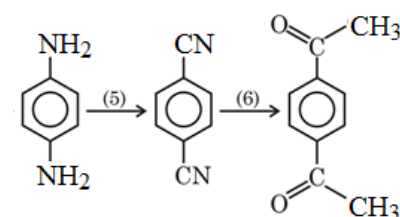
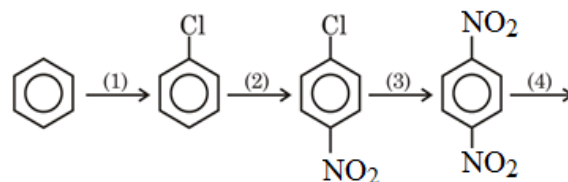
(C) CCl_3CHO gives CHCl_3 , not iodoform.

(D) Electronic effects

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (116.64)**



Accounting for yield, moles of product formed

$$= 0.90 \times 0.80 \times 1$$

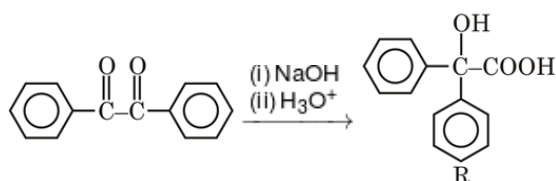
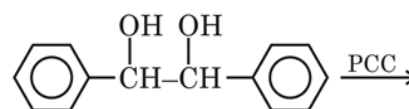
$$= 0.72$$

$$\text{molar mass} = 162 \text{ g}$$

$$\text{mass of product obtained} = 0.72 \times 162$$

$$= 116.64 \text{ g}$$

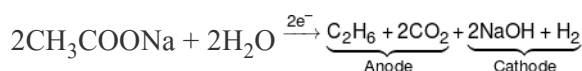
2. **Ans (25.33)**



$$\text{Molar mass} = 228 \text{ g/mol}$$

$$\text{DoU} = 9$$

3. Ans (3.00)



Number of Faradays passed

$$= \frac{9.65 \times 10^{-3} \times 2500}{96500} = 2.5 \times 10^{-4}$$

= number of eq. of C_2H_6 produced.

$$\text{Equivalent mass of } \text{C}_2\text{H}_6 = \frac{30}{2} = 15 \text{ g. eq.}$$

∴ Mass in mg of C_2H_6 (ideally)

$$= 2.5 \times 10^{-4} \times 15 \times 10^3 \text{ mg}$$

As current efficiency is 80%,

$$\begin{aligned} \text{mass of } \text{C}_2\text{H}_6 \text{ produced} &= \frac{80}{100} \times 3.75 \\ &= 3.00 \text{ mg} \end{aligned}$$

4. Ans (10.00)

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5. Ans (1.50)

a = 2 (Aspartic acid, Glutamic acid)

b = 3 (Arginine, Histidine, Lysine)

c = 4 (Phenylalanine, Tyrosine, Tryptophan, Histidine)

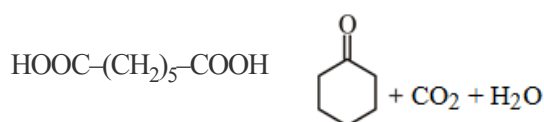
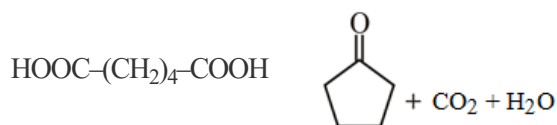
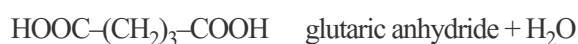
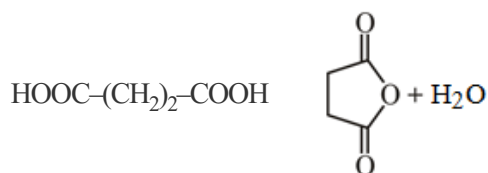
PART-2 : CHEMISTRY

SECTION-III

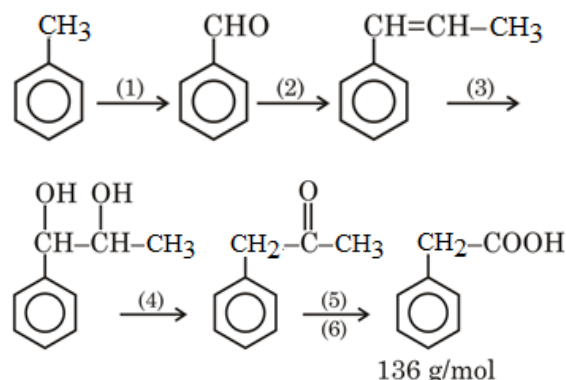
1. Ans (5)

Reactant

Product



2. Ans (8)



3. Ans (2)

Penicillin, ofloxacin-bactericidal

Erythromycin, tetracycline, chloramphenicol-bacteriostatic.

4. Ans (6)

Polyethylene, Teflon, PVC, PMMA, Polystyrene, Natural rubber,

5. Ans (4)

(iv), (v), (vi), (vii)

6. Ans (3)

$$R_f = \frac{9}{15} = 0.6$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

The possible powers of 'x' in first 3 brackets are

0, 2, 4, 6 ;

$$0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, 2, \frac{7}{3} ;$$

$$0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}$$

To get x^6 we need sum of powers

$$\begin{aligned} &\left(4, \frac{2}{3}, \frac{4}{3}\right), (4, 1, 1), \left(4, \frac{4}{3}, \frac{2}{3}\right), \left(4, \frac{5}{3}, \frac{1}{3}\right), (4, 2, 0), (6, 0, 0) \\ &{}^3\text{C}_2 \cdot {}^7\text{C}_2 \cdot {}^4\text{C}_4 + {}^3\text{C}_2 \cdot (-{}^7\text{C}_3) \cdot {}^4\text{C}_3 + {}^3\text{C}_2 \cdot {}^7\text{C}_4 \cdot {}^4\text{C}_2 \\ &+ {}^3\text{C}_2 \cdot (-{}^7\text{C}_2) \cdot {}^4\text{C}_1 + {}^3\text{C}_2 \cdot {}^7\text{C}_6 \cdot {}^4\text{C}_0 + {}^3\text{C}_3 = 43 \end{aligned}$$

2. Ans (C)

$$x_1 \quad G_1 \quad x_2 \quad G_2 \quad x_3 \quad G_3 \quad x_4 \quad G_4 \quad x_5 \quad G_5 \quad x_6$$

$$x_1 + \dots + x_6 = 15, \quad x_1, x_6 \geq 0, \quad x_2, x_3, x_4, x_5 \geq 2$$

$$\therefore {}^{7+6-1}\text{C}_{6-1} = {}^{12}\text{C}_5$$

$$\therefore P = \frac{{}^{12}\text{C}_5 \cdot 5! \cdot 15!}{20!} = \frac{12! \cdot 15!}{7! \cdot 20!}$$

3. Ans (A,C)

$$\begin{aligned} & \frac{1}{2^{15}} \sum_{r=0}^{15} (-1)^r 5^{15-r} \frac{30!}{r! (30-r)!} \frac{(30-r)!}{(15-r)!} 15! \\ &= \left(\frac{5}{2}\right)^{15} \sum_{r=0}^{15} \left(-\frac{1}{5}\right)^r \frac{30!}{15! 15!} \frac{15!}{r! (15-r)!} \\ &= \left(\frac{5}{2}\right)^{15} {}^{30}C_{15} \sum_{r=0}^{15} {}^{15}C_r \left(-\frac{1}{5}\right)^r \\ &= \left(\frac{5}{2}\right)^{15} {}^{30}C_{15} \left(\frac{4}{5}\right)^{15} = {}^{30}C_{15} 2^{15} \end{aligned}$$

4. Ans (B,D)

Clearly '0' can't be any of the digit

Suppose 1 occurs x_1 times

2 occurs x_2 times

$\vdots$

9 occurs x_9 times

$$\therefore x_1 + x_2 + \dots + x_9 = 15$$

$$\therefore \text{No of integral sol} = {}^{15+9-1}C_{9-1} = {}^{23}C_8$$

$$\begin{aligned} &= \frac{23!}{8! 15!} \\ &= 23 \cdot 22 \cdot 3 \cdot 19 \cdot 17 \end{aligned}$$

5. Ans (B,D)

$$\begin{aligned} & \sqrt{6} - \sqrt{5} > \sqrt{8} - \sqrt{7} > \log_{\sqrt{6}-\sqrt{5}} \sqrt{8} - \sqrt{7} < 1 \\ & \therefore \sqrt{6} - \sqrt{5} \in (0, 1) \end{aligned}$$

Also

$$\begin{aligned} & (\sqrt[3]{2} + \sqrt[3]{5}) \left((\sqrt[3]{2})^2 + (\sqrt[3]{5})^2 - \sqrt[3]{2} \cdot \sqrt[3]{5} \right) \\ &= 2 + 5 = 7 \end{aligned}$$

$$\therefore \log_7^7 = 1$$

$$T_{r+1} = {}^{105}C_r 7^{\frac{105-r}{7}} \cdot 2^{r/5} = {}^{105}C_r 7^{15} \cdot \frac{2^{r/5}}{7^{r/7}}$$

6. Ans (A,C)

$$S = \sum_{r=1}^{2023} \sin(rx) \cos(2024-r)x$$

$$S = \sum_{r=1}^{2023} \sin(2024-r)x \cdot \cos rx$$

$$2S = \sum_{r=1}^{2023} \sin 2024x$$

$$S = \frac{2023}{2} \sin 2024x = \frac{2023}{2}$$

$$\sin(2024x) = 1$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.05)

$$\begin{aligned} & \sum_{n=4}^{2023} \frac{n+3}{(n+1)! (n+3)^2} = \sum_{n=4}^{2023} \frac{1}{(n+1)! (n+3)} \\ &= \sum_{n=4}^{2023} \frac{n+2}{(n+3)!} = \sum_{n=4}^{2023} \frac{(n+3)-1}{(n+3)!} \\ &= \sum_{n=4}^{2023} \left(\frac{1}{(n+2)!} - \frac{1}{(n+3)!} \right) \\ &= \frac{1}{6!} - \frac{1}{2026!} \end{aligned}$$

2. Ans (2.00)

$$\text{Let } \log_{\sqrt{2}-1} \tan x = t$$

$$t \cdot \sqrt{\frac{3}{2} + \frac{1}{t}} = -1 \quad \therefore t < 0$$

$$\Rightarrow \frac{3}{2} + \frac{1}{t} = \frac{1}{t^2} \Rightarrow 3t^2 + 2t = 2$$

$$\Rightarrow 3t^2 + 2t - 2 = 0 \Rightarrow t = \frac{-2 \pm \sqrt{28}}{6}$$

$$t = \frac{-1 \pm \sqrt{7}}{3} \quad \& \quad \frac{3}{2} + \frac{1}{t} > 0$$

$$\Rightarrow \frac{3}{2}t + 1 < 0 \Rightarrow t < -\frac{2}{3}$$

$$\therefore t = \frac{-1 - \sqrt{7}}{3}; \log_{\sqrt{2}-1} \tan x = \frac{-(\sqrt{7}+1)}{3}$$

$$\tan x = (\sqrt{2}-1)^{-\left(\frac{\sqrt{7}+1}{3}\right)} = \frac{1}{(0.414)^{\sqrt{7}+1}}$$

3. Ans (0.00)

$$10 - 2\sin^2\theta \cdot \cos^2\theta - 6\sin\theta \cos\theta (\sin\theta + \cos\theta) = 0$$

$$(s^4 + c^4) + 9(s^2 + c^2) - 6s^2c - 6c^2s = 0 \quad (s$$

$$= \sin\theta \text{ and } c = \cos\theta)$$

$$(s^2 - 3c)^2 + (c^2 - 3s)^2 = 0$$

$$\therefore \sin^2\theta = 3\cos\theta \quad \text{and} \quad \cos^2\theta = 3\sin\theta$$

$$\therefore \text{No solution}$$

4. Ans (6.00)

$$\frac{1}{\log_{pq}^q} - \frac{1}{\log_{pq}^p} = \log_{pq}^{pq} - \log_{pq}^{pq}$$

$$= \log_q^p - \log_p^q$$

$$= \sqrt{(\log_q^p + \log_p^q)^2 - 4} = 45$$

$$\operatorname{cosec} \theta = 45 \text{ has 6 solutions}$$

5. Ans (19.00)

$$\frac{m}{n} = \frac{\frac{7}{11} \cdot \frac{6}{10} \cdot \frac{5}{9} \cdot \frac{4}{8}}{\frac{7}{11} \cdot \frac{4}{10} + \frac{7}{11} \cdot \frac{6}{10} \cdot \frac{5}{9} \cdot \frac{4}{8} + \frac{7}{11} \cdot \frac{6}{10} \cdot \frac{5}{9} \cdot \frac{4}{8} \cdot \frac{3}{7} \cdot \frac{4}{6} + \frac{7}{11} \cdot \frac{6}{10} \cdot \frac{5}{9} \cdot \frac{4}{8} \cdot \frac{3}{7} \cdot \frac{4}{6} \cdot \frac{2}{5} \cdot \frac{1}{1}}$$

$$= \frac{7}{26}$$

$$\therefore n-m = 19$$

6. Ans (0.36)

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{7\}$$

$$C = \{7, 9, 3, 1\}$$

$$(A \cap B) \cup C = \{7, 9, 3, 1\}$$

$$\therefore \text{Required probability} = 0.11 + 0.14 + 0.03 + 0.08 = 0.36$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (2)

$$X ; n_1 p_1 = 30 ; n_1 p_1 q_1 = 5$$

$$\therefore q_1 = \frac{1}{6}, p_1 = \frac{5}{6}, n_1 = 36$$

$$Y : n_2 p_2 = 9 ; n_2 p_2 q_2 = \frac{27}{4}$$

$$\therefore q_2 = \frac{3}{4}, p_2 = \frac{1}{4}, n_2 = 36$$

$$\text{Now } a^{a+b} = b^{12} \Rightarrow b = a^{\frac{a+b}{12}}$$

$$\therefore b^{a+b} = a^3 \Rightarrow a^{\frac{(a+b)^2}{12}} = a^3,$$

$$\therefore a + b = 6 \text{ and } b^2 = a$$

$$\therefore b^2 + b = 6 \Rightarrow b = 2, a = 4$$

$$\frac{P_1(X=4)}{P_2(Y=2)} = \frac{{}^{36}C_4 \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right)^{32}}{{}^{36}C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{34}} = \frac{11 \cdot 17 \cdot 5^4 \cdot 2^{35}}{3^{70}}$$

$$\therefore p = 5, q = 3$$

2. Ans (5)

$$\sec \theta + \operatorname{cosec} \theta = \sin \theta + \cos \theta$$

$$\Rightarrow (\sin \theta + \cos \theta) (1 - \sin \theta \cos \theta) = 0$$

$$\therefore \tan \theta = -1 \text{ on } \sin 2\theta = 2$$

$$\therefore \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$A = \left\{ \frac{3\pi}{4}, \frac{7\pi}{4} \right\}$$

$$\sqrt{1 - \sin \theta} = \cos \theta, \text{ squaring}$$

$$\Rightarrow \sin \theta = 0, 1$$

$$\theta = 0, \pi, 2\pi, 3\pi, \frac{\pi}{2}, \frac{5\pi}{2}$$

putting and verifying

$$B = \left\{ 0, 2\pi, \frac{\pi}{2}, \frac{5\pi}{2} \right\}$$

$$A \cup B = \left\{ 0, \frac{\pi}{2}, \frac{3\pi}{4}, 2\pi, \frac{7\pi}{4}, \frac{5\pi}{2} \right\}$$

$$\theta = \frac{3\pi}{4} \text{ doesn't satisfy } C$$

$$\therefore n(A \cup B) \cap C = 5$$

3. Ans (3)

$$3 + 3 \log_2 \cos 6x + 3 \log_2 \cos x = \log_2 (\cos^3 5x + \cos^3 7x)$$

$$\Rightarrow \log_2 (8 \cos^3 6x \cdot \cos^3 x) = \log_2 (\cos^3 5x + \cos^3 7x)$$

$$\Rightarrow 8 \cos^3 6x \cdot \cos^3 x = \cos^3 5x + \cos^3 7x$$

$$\Rightarrow (2 \cos 6x \cdot \cos x)^3 = \cos^3 5x + \cos^3 7x$$

$$\Rightarrow (\cos 5x + \cos 7x)^3 = \cos^3 5x + \cos^3 7x$$

$$(a + b)^3 = a^3 + b^3$$

$$\Rightarrow 3ab(a + b) = 0$$

$$\therefore a + b \neq 0 \quad \therefore a = 0 \text{ or } b = 0$$

$$\text{So } \cos 5x = 0$$

$$\Rightarrow x = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2} \text{ but } \cos 7x, \cos 6x, \cos x > 0$$

$$\Rightarrow x = 3\pi/10$$

$$\text{or } \cos 7x = 0$$

$$x = \frac{\pi}{14}, \frac{3\pi}{14}, \frac{5\pi}{14}, \frac{\pi}{2} \text{ but } \cos 5x, \cos 6x, \cos x > 0$$

$$\Rightarrow \frac{\pi}{14}, \frac{5\pi}{14}$$

4. Ans (2)

$$\sum x_i = 100 + 70 = 170$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$\sum x_i^2 = \left(\frac{5}{2} + p \right)^2 \cdot 20 + \left(\frac{7}{2} \right)^2 \cdot 20 + \left(\frac{5}{2} - p \right)^2 \cdot 20$$

$$= 20 \left[\frac{99}{4} + 2p^2 \right]$$

$$\sum x_i = 20 \left[\frac{5}{2} + p + \frac{7}{2} + \frac{5}{2} - p \right]$$

$$= 20 \left[5 + \frac{7}{2} \right] = 170$$

$$\frac{4}{9} = 20 \frac{\left[\frac{99}{4} + 2p^2 \right]}{60} - \frac{289}{36}$$

$$\frac{305 - 297}{12} = 2p^2$$

$$\therefore 6p^2 = 2$$

5. Ans (6)

$$f(m) = \sum_{r=1}^m \frac{(m^r + m^{r-1}) \left(1 - \frac{1}{r}\right)}{{}^m C_r}$$

$$= \sum_{r=1}^m \frac{m^{r-1} (m+1)(r-1)}{r \cdot {}^m C_r}$$

$$= \sum_{r=1}^m \frac{m^r}{{}^m C_r} + \frac{m^{r-1} (r-1) - m^r}{r \cdot {}^m C_r}$$

$$= \sum_{r=1}^m \frac{m^r}{{}^m C_r} - \frac{m^{r-1} (m-r+1)}{r \cdot m!} r! (m-r)!$$

$$= \sum_{r=1}^m \frac{m^r}{{}^m C_r} - \frac{m^{r-1} (m-r+1)}{r \cdot {}^m C_r} (m - (r-1))$$

$$= \sum_{r=1}^m \frac{m^r}{{}^m C_r} - \frac{m^{r-1} (m-r+1)}{r \cdot m!} r! (m-r)!$$

$$= \sum_{r=1}^m \frac{m^r}{{}^m C_r} - \frac{m^{r-1} r(r-1)!(m - (r-1))!}{r \cdot m!}$$

$$= \sum_{r=1}^m \left(\frac{m^r}{{}^m C_r} - \frac{m^{r-1}}{{}^m C_{r-1}} \right)$$

$$= m^m - 1$$

6. Ans (4)

R G B Y P (These pattern should not occur).

The possible combination i.e

B P G Y R

B P R Y G

B R P G Y

B R Y G P

G P B R Y

G Y R B P

G Y R P B

P B R Y G

P G Y R B

R B P G Y

R Y G P B

Y G P B R

Y G P R B

Y R B P G

$$\frac{2}{7}N = \frac{2}{7} \cdot 14 = 4$$